5.24. The lifetimes of interactive computer chips produced by a certain semiconductor manufacturer are normally distributed with parameters $\mu = 1.4 \times 10^6$ hours and $\sigma = 3 \times 10^5$ hours. What is the approximate probability that a batch of 100 chips will contain at least 20 whose lifetimes are less than 1.8×10^6?

5.34. Jones figures that the total number of thousands of miles that an auto can be driven before it would need to be junked is an exponential random variable with parameter $\frac{1}{20}$. Smith has a used car that he claims has been driven only 10,000 miles. If Jones purchases the car, what is the probability that she would get at least 20,000 additional miles out of it? Repeat under the assumption that the lifetime mileage of the car is not exponentially distributed,

but rather is (in thousands of miles) uniformly distributed over $(0, 40)$.

- Suppose the current measurements in a strip of wire are assumed to follow a normal distribution with a mean of 10 milliamperes and a variance of 4 milliamperes.

  (a) What is the probability that a measurement will exceed 13 milliamperes?

  (b) Determine the value for which the probability that a current measurement is below this value is 0.98.

In a large corporate computer network, user log-ons to the system can be modeled as a Poisson process with $\lambda = 25$ log-ons per hour. This means that if we let X denote the time in hours from the start of the interval until the first log-on, then X has an exponential distribution with $\lambda = 25$.

(a) What is the probability that the time until the next log-on exceeds 6 minutes?

(b) Determine the length of time t such that the probability that the time until the next log-on exceeds t is 0.90.